
AI governance around the world

Country profile: European Union

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About The Alan Turing Institute

The Alan Turing Institute is the UK's national institute for data science and artificial intelligence.

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About the project

Introduction

The international AI governance landscape continues to evolve as jurisdictions around the world balance the need to navigate increased competition with earlier commitments to advancing international alignment and cooperation on AI governance and safety. Many jurisdictions have recognised the strategic implications of AI, both for economic prosperity and national security, and accelerated investment in domestic AI infrastructure has begun to engender a more competitive international environment. However, the role of international trade and global cooperation remains crucial in realising the economic benefits of AI while effectively managing the technology's risks.

As high-level AI principles are translated into concrete policies and distinct regulatory frameworks, jurisdictions are managing the tensions between competitive and cooperative priorities. Meaningful progress towards effective international AI governance will require a clear and detailed understanding of how different countries are approaching AI governance in practice. Tracking primary source policy initiatives over the past decade, the *AI governance around the world* project sheds light on shifting national priorities on AI, dynamics of cooperation and competition, and potential areas of international alignment.

Project aims

The *AI governance around the world* project seeks to map this evolving landscape through a series of country profiles based on a consistent framework. Each profile provides a descriptive overview of a jurisdiction's approach to AI regulation and standardisation, highlighting the high-level aims and principles, definitions of relevant technologies, key policy initiatives and the main features of the respective standardisation systems. Drawing exclusively on primary sources, the country profiles analyse legal and policy instruments, national strategies, standardisation initiatives, and public investments that reflect the varied and evolving approaches that different jurisdictions are taking to AI governance.

This series offers a foundation for comparative analysis and future work on global regulatory interoperability without commenting on the efficacy of the specific governance models being adopted at the jurisdictional level. As more profiles are developed, the project aims to contribute to international understanding of where alignment is emerging and where deeper coordination may be needed.



European Union

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Executive summary



The European Union (EU) is taking a ‘harder’ AI governance approach than most other jurisdictions, setting up legally binding requirements to ensure AI technologies are developed and used in a safe and responsible way, minimising the risk of harm to the health, safety, or fundamental rights of EU citizens. The rules for AI products and services that the EU is developing will have implications for companies well beyond the Union’s borders. They will apply to products and services that are commercialised within the European market as well as to those that, even if not introduced in the EU market, have an impact on physical persons residing in the EU.

Standards are set to play a central role in the EU’s regulatory framework for AI technologies. Developed by European Standards Organisations (ESOs) based on a request by the European Commission, European harmonised standards will provide the technical specifications necessary to support the implementation of the *EU AI Act*. Like most standards, the adoption of European harmonised standards is voluntary. However, if adopted, they will grant companies a presumption of conformity with specific *EU AI Act* requirements.

Key policy initiatives

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- A vertical timeline on the left side of the page, marked with orange circles, indicates the progression of time from 2018 to 2025. Each event is represented by a horizontal orange bar with a date on the left, a title in bold, and a description below it.
- Apr 2018** ● **Artificial Intelligence for Europe**
First European AI Strategy
 - Apr 2019** ● **Ethics Guidelines for Trustworthy AI**
Voluntary framework defining values for responsible AI
 - Apr 2021** ● **Proposal for the EU AI Act**
First draft proposal for a regulation laying down harmonised rules on AI
 - Feb 2024** ● **Creation of the EU AI Office**
Creation of the EU AI Office, the primary body responsible for the implementation of the EU AI Act
 - Aug 2024** ● **The EU AI Act enters into force**
The EU AI Act entered into force on 1 August 2024
 - Sep 2024** ● **AI Pact**
Voluntary pledge by companies to start applying the principles of the EU AI Act before they become enforceable
 - April 2025** ● **AI Continent Action Plan**
Comprehensive AI strategy with the objective to position the EU as a global leader in AI
 - Aug 2025** ● **GPAI Code of practice**
Voluntary framework to help providers of GPAI models comply with Artt. 53 and 55 of the EU AI Act

Regulatory approach to AI

High-level aims and principles

In *Artificial Intelligence for Europe*, the EU's first AI strategy, the European Commission set three overarching objectives: boost AI uptake across the economy to strengthen the EU's technological and industrial capacity; address the socio-economic challenges brought about by AI technologies; and develop an appropriate ethical and legal framework to enable the development of trustworthy AI.¹

These broad strategic objectives are reflected in the *EU AI Act*, which aims to ensure the proper functioning of the single market by creating the conditions for the development and use of trustworthy artificial intelligence in the Union. This includes developing an effective governance infrastructure that can support the development and use of lawful, safe, and trustworthy AI systems while also ensuring these technologies can contribute to innovation and economic growth.

The EU's approach to governing AI technologies builds on a set of ethical principles first presented by the EU's High Level Expert Group (HLEG) in its *Ethics Guidelines for Trustworthy AI*² and later integrated in the *EU AI Act*. These include data governance and quality, traceability and technical documentation, transparency and provision of information, human oversight and accuracy, robustness, and security. These principles also provide the foundation of the ten standardisation deliverables that the European Committee for Standardisation (CEN) and the European Committee for Electrotechnical Standardisation (CENELEC) have been tasked with developing, which providers and users of AI systems will be able to use to demonstrate compliance with the *EU AI Act*'s requirements.³

In April 2025, the European Commission published a new strategy to accelerate the development and adoption of AI throughout the EU economy. The AI Continent Action

¹ European Commission, 'Artificial Intelligence for Europe', April 2018, <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=COM:2018:237:FIN>.

² High-Level Expert Group on AI, 'Ethics guidelines for Trustworthy AI', April 2019, <https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai>.

³ European Commission, 'Commission Implementing Decision on a standardisation request to the European Committee for Standardisation and the European Committee for Electrotechnical Standardisation in support of Union policy on artificial intelligence', May 2023, [https://ec.europa.eu/transparency/documents-register/detail?ref=C\(2023\)3215&lang=en](https://ec.europa.eu/transparency/documents-register/detail?ref=C(2023)3215&lang=en).

Plan⁴ focuses on five areas of intervention: a) building large-scale AI infrastructure; 2) increasing access to high-quality data; 3) supporting AI adoption in strategic industry sectors; 4) strengthening AI skills and attracting AI talent; 5) simplifying the implementation of the EU AI Act. The EU's new AI strategy remains largely consistent with the ethical values and principles put forth in previous policy initiatives while re-focusing the Commission's policy priorities towards technical research, industry applications, and AI infrastructure.

Definitions of relevant technologies

Based on Art. 3(1) of the *EU AI Act*, an AI system is “a machine-based system designed to operate with varying levels of autonomy and that may exhibit adaptiveness after deployment and that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments.”⁵ Compared to the definition proposed by the European Commission in the first draft of the *EU AI Act* in 2021,⁶ the definition adopted in the final text of the regulation is more streamlined and specific, explicitly adding the elements of autonomy and adaptiveness as distinctive features of AI systems. This definition also aligns with the definition of ‘AI system’ provided by the OECD.⁷

In February 2025, the European Commission published a set of guidelines to clarify and support the practical application of the legal definition of ‘AI system’ under Art. 3(1) of the *EU AI Act*. These guidelines provided concrete examples of technologies that fall under the scope of the regulation and helped clarify some areas of ambiguity. Nevertheless, some concerns have been raised that parts of the guidelines may effectively narrow the scope Art. 3(1), particularly those introducing exemptions for ‘simple prediction systems’ and ‘systems improving mathematical optimisation.’

Key policy initiatives

The EU has been an early mover in AI regulation. The centrepiece of the European Commission's approach is the *Artificial Intelligence Act (EU AI Act)*, a comprehensive

⁴ European Commission, ‘AI Continent Action Plan’, April 2025, https://commission.europa.eu/topics/eu-competitiveness/ai-continent_en.

⁵ Regulation (EU) 2024/1689.

⁶ Based on Art 3 of the original proposal, “‘artificial intelligence system’ (AI system) means software that is developed with one or more of the techniques and approaches listed in Annex I and can, for a given set of human-defined objectives, generate outputs such as content, predictions, recommendations, or decisions influencing the environments they interact with.” Source: [COM\(2021\)0206](COM(2021)0206).

⁷ OECD, ‘Updates to the OECD’s definition of an AI system explained’, November 2023, <https://oecd.ai/en/work/ai-system-definition-update>.

regulation laying out rules to ensure the proper functioning of the single European market for AI. The EU's broader digital regulatory ecosystem also includes other laws that influence the use and commercialisation of AI technologies. These include the *Data Act*, the *Data Governance Act*, the *Digital Market Act*, the *Digital Services Act*, the *General Data Protection Regulation*, and the *Product Liability Directive*. While these laws are not AI-specific and, as such, will not be analysed in detail here, it is important to acknowledge their role in European Commission's overall approach to AI governance.

Several new initiatives are also planned for the second half of 2025 and into 2026. By the end of 2025, the European Commission is set publish two new strategic documents: the *Apply AI Strategy*, which will focus on leveraging AI's potential in target sectors and for the delivery of public services; and the *European Strategy for AI in Science*, which will promote the responsible use of AI in research and innovation. In addition, the European Commission is expected to introduce the *Cloud and AI Development Act* by the first quarter of 2026—a regulation aimed at incentivising investment in cloud and edge capacity.

Horizontal initiatives

Artificial Intelligence Act

The *EU AI Act*⁸ is the flagship EU regulation on AI technologies. Entered into force on 1 August 2024, the Act lays out a risk-based regulatory framework for the development, placing on the market, and use of AI technologies in the EU.

The *EU AI Act* includes provisions regulating both AI systems and general purpose AI models (GPAI). AI systems are subject to proportionally stringent requirements based on the level of risk related to their 'intended use', meaning the use for which they are intended by providers. Uses of AI systems that pose an *unacceptable risk* of violating fundamental rights and contravening EU values are prohibited. These include AI applications that can cause significant harm to persons or groups of persons by manipulating someone's behaviour; exploiting the vulnerabilities of a group of persons based on their age, physical or mental disability, or socioeconomic status; scoring persons based on their social behaviour; predicting the risk of persons committing a crime based solely on their personality traits; inferring a person's emotions on the workplace or in education institutions; identifying people in real-time through the analysis of biometric data in public spaces. AI applications that pose a *high risk*, meaning a significant risk to the health, safety or fundamental rights of individuals, have

⁸ Regulation (EU) 2024/1689.

to comply with a stringent set of requirements. These range from deploying a risk management system and data management processes, to enabling sufficient transparency, ensuring accuracy, robustness, and cybersecurity, ensuring human oversight, and automatic record-keeping. Providers of high-risk systems will also be required to establish a quality management system through which they can demonstrate the compliance of their systems with such requirements. Only AI systems whose compliance with these requirements is certified can be commercialised in the EU market. The AI act also introduces obligations for AI applications that pose a specific *transparency risk*. For example, it is mandatory to ensure natural persons are aware that they are interacting directly with an AI system or consuming synthetically generated content (e.g., images, text, audio, video, etc.). Finally, AI systems intended for any other use are considered to generate only a *minimal risk* and are not subject to mandatory requirements.

The *EU AI Act* also lays out obligations for GPAI model providers. These include sharing specific types of information as part of the systems' technical documentation and ensuring that downstream providers and deployers have the necessary information and support to use the system in compliance with the *EU AI Act*. More stringent obligations exist for providers of GPAI models that can pose systemic risks—i.e., models that can have a significant impact on the Union market due to their reach, or due to actual or reasonably foreseeable negative effects on public health, safety, public security, fundamental rights, or the society as a whole, that can be propagated at scale across the value chain. Providers of GPAI models with systemic risk shall perform model evaluations in accordance with standardised protocols and tools, assess and mitigate possible systemic risks at the Union level, report serious incidents to the AI Office, and ensure adequate levels of cybersecurity protection for the models and their infrastructure.

Crucially, the *EU AI Act* will have relevant implications for organisations even beyond the EU's borders. Providers that commercialise AI systems in the EU market will fall under the scope of the regulation irrespective of their place of establishment. And even for deployers—which are otherwise affected by the *EU AI Act*'s obligations only when established within the EU—the regulation will apply if the AI systems that they use can affect natural persons residing in the EU. As such, the scope of application of the AI Act is admittedly very large. Whether the regulation will be implemented as broadly as it potentially can, however, remains an open question.

A key challenge for the success of the *EU AI Act* is how well the regulation will be implemented and enforced. This will require a combination of legal and technical skills, as well as substantive efforts from the EU institutions and the various national bodies responsible for enforcing the regulation. An array of new institutional bodies has been

tasked with various implementation and enforcement responsibilities. At the Union level, the European Commission and its AI Office are responsible for most implementation and enforcement functions. They are supported in this role by the AI Board, which represents Member States, the Advisory Forum, which brings together various stakeholder groups to offer technical expertise and support, and the Scientific Panel of Independent Experts, which will advise the European Commission and national authorities on systemic risks, model classification and evaluation methodologies for GPAI models. The AI Office is also setting up the *AI Act Service Desk* to help stakeholders understand what obligations they are subject to and help them comply, including by responding to *ad hoc* queries. At the national level, Member States are required to designate national competent authorities, which include at least one notifying authority and one market surveillance authority.

Different bodies are also developing resources to support and facilitate compliance with the *EU AI Act*. For instance, European Standards Organisations are developing standards to specify the technical requirements needed to comply with Section 2 of the *EU AI Act*; the European Commission's AI Office led a multistakeholder initiative to draft a *General Purpose AI Code of Practice*, which was adopted in August 2025 and supports providers of GPAI models to comply with the relevant *EU AI Act* requirements; the European Commission has also been publishing a series of guidelines and delegated and implementing acts to facilitate the interpretation and application of specific *EU AI Act* provisions. In February 2025, the Commission published two sets of guidelines to clarify the definition of 'AI system' and specify what AI practices shall be considered as prohibited under Art. 5 of the *EU AI Act*.⁹ More recently, it published guidelines on the scope of the obligations for GPAI model providers.¹⁰

General Purpose AI Code of Practice

The General-Purpose AI Code of Practice was developed to help providers of GPAI models comply with requirements on safety, transparency and copyright set by Artt. 53 and 55 of the *EU AI Act*, which became applicable on 2 August 2025. Signatories of the Code will be able to demonstrate compliance with the relevant provisions of the *EU AI Act* by adhering to the voluntary measures described in the Code. Organisations that refuse to sign it, instead, will be required to demonstrate compliance through alternative and likely more burdensome processes.

⁹ European Commission, Guidelines on prohibited artificial intelligence practices established by Regulation (EU) 2024/1689 (AI Act)', February 2025, <https://digital-strategy.ec.europa.eu/en/library/commission-publishes-guidelines-prohibited-artificial-intelligence-ai-practices-defined-ai-act>.

¹⁰ European Commission, 'Guidelines on the scope of obligations for providers of general-purpose AI models under the AI Act', July 2025, <https://digital-strategy.ec.europa.eu/en/library/guidelines-scope-obligations-providers-general-purpose-ai-models-under-ai-act>.

Published on 10 July 2025 and approved by the Commission on 1 August 2025¹¹, the Code was prepared by a team of independent experts through a multi-stakeholder process bringing together more than 1000 organisations from industry, civil society, and the scientific community. The Code consists of three chapters: Transparency, Copyright, and Safety and Security. The Transparency Chapter offers a form that GPAI providers can use to share information that should be made available to the AI Office, national competent authorities, and downstream providers. The Copyright Chapter provides guidance on how to comply with EU law on copyright and IPs. Differently from the first two chapters, which apply to all GPAI model providers, the final chapter on Safety and Security only applies to a small subset of organisations that provide GPAI models with systemic risk under Art. 55. The third chapter details a series of measures that providers of GPAI models with systemic risk shall adopt to identify, measure, and control systemic risks. Finally, a guiding principle cutting across all three chapters is the importance of ethical AI development. In this sense, GPAI providers are encouraged to reflect on the ethical impact of their models and to ensure that their use aligns with broader societal norms and values.

At the time of writing, some of the main organisations that signed up to the Code include OpenAI, Google, Microsoft, Anthropic, Mistral AI, IBM, Amazon, Cohere, and Aleph Alpha. xAI only signed up to the Safety and Security Chapter, implying it will demonstrate compliance with the remaining obligations of the EU AI Act via alternative means, while Meta officially refused to sign it.¹²

EU AI Pact

Soon after the entry into force of the EU AI Act, the European Commission published the EU AI Pact¹³, a voluntary pledge by multinational corporations and European SMEs to start applying the principles of the EU AI Act before they become enforceable. Signatories to the EU AI Pact committed to developing and implementing an AI governance strategy for AI adoption within their organisations and starting to work towards compliance with the EU AI Act; mapping systems that are likely to fall under the definition of 'high-risk' AI systems under the EU AI Act; and promoting AI literacy within their organisations. More than half the signatories also committed to additional actions

¹¹ European Commission, 'Commission Opinion on the assessment of the General-Purpose AI Code of Practice', August 2024, <https://digital-strategy.ec.europa.eu/en/library/commission-opinion-assessment-general-purpose-ai-code-practice>.

¹² *Ibid*.

¹³ European Commission, 'EU AI Pact', September 2024, <https://digital-strategy.ec.europa.eu/en/policies/ai-pact>.

on requirements introduced by the AI Act concerning high-risk AI systems and transparency obligations for GPAI providers.

Data-related legislation

The EU has multiple legislative instruments relating to data governance, data protection, and data sharing that will impact the development and deployment of AI systems. These include the *General Data Protection Regulation (GDPR)*, the *Data Governance Act*, the *Data Act* and the *Open Data Directive*.

The *GDPR* is described by the EU institutions as “the strongest privacy and security law in the world”,¹⁴ and defines individuals’ fundamental rights, the obligations of those processing data, methods for ensuring compliance and sanctions for those in breach of the rules. The *Data Governance Act*, *Data Act* and *Open Data Directive* aim to make more data available for use and strengthen data sharing mechanisms, widening the scope of AI systems that can be developed.

Vertical initiatives

While the *EU AI Act* should be primarily viewed as a horizontal piece of legislation, it contains many sector-specific provisions. These aim to ensure consistency between the AI Act and existing sectoral legislation and, in some cases, provide new requirements based on the unique risks of deploying AI systems in specific sectors. The clearest example of this vertical approach is the classification of AI systems as ‘high-risk’ when these are intended to be used in ways that pose a risk of harm to health, safety or fundamental rights in specific sectors, such as education and vocational training, critical infrastructure management, or biometric identification of physical persons.

Approach to AI standardisation

Main features of the standardisation system

European standardisation is a unique system: there are standardisation structures and processes that operate at both national and regional (European) levels, and around 30% of European standards are adopted under the guidance of the European Commission, giving them a special status in the EU legal system.

¹⁴ Council of the EU and the European Council, ‘The general data protection regulation’, 2023, <https://www.consilium.europa.eu/en/policies/data-protection/data-protection-regulation/>.

The Commission has the twofold power to set the strategic priorities of European standardisation through the annual *Union work programme* and initiate the process that leads to developing European harmonised standards, i.e. European standards “adopted on the basis of a request made by the Commission for the application of Union harmonisation legislation”.¹⁵ Harmonised standards differ from other European standards due to their distinctive legislative function in providing technical specifications to support the implementation of EU legislation. While harmonised standards are technically voluntary, their adoption grants a presumption of legal conformity with related EU legislation. Additionally, when a harmonised standard is adopted, National Standards Bodies must withdraw any conflicting national standard within a “reasonable deadline”.¹⁶

The rationale behind the Commission’s direct involvement in European standardisation processes lies in the key role that standards play in harmonising EU legislation across Member States. For instance, harmonised standards are instrumental to the construction of a healthy internal market: the system of harmonised standards was first established as a way to specify quality and safety requirements of products and services traded within the EU. Additionally, standards can play a role in fostering the development of European businesses, providing value to consumers by enhancing product quality, interoperability and the circulation of information, and supporting international trade.

As already mentioned, the Commission strictly collaborates with ESOs. These are CEN, CENELEC, and the European Telecommunications Standards Institute (ETSI). They are membership-based associations officially recognised by Regulation (EU) No 1025/2012 as the providers of European standards in specific sectors.

In addition to working under the Commission’s guidance to support the implementation of European legislation, ESOs also develop European standards that are requested by their own members. Membership in CEN and CENELEC is restricted to National Standards Bodies. Stakeholders who wish to participate in their activities can only do so through the respective National Standards Bodies. ETSI, instead, is open to the direct participation of industry organisations.

¹⁵ Regulation (EU) No 1025/2012, Art. 2.

¹⁶ *Ibid.*, Art. 3.

National Standards Bodies are the one-stop shop for all stakeholders interested in partaking in either European or International standardisation. These include industry associations, businesses, academia, and civil society representatives.

European AI-focused standardisation activities

EU standardisation for AI technologies is carried out by a Joint Technical Committee (CEN-CENELEC/JTC 21) established following the publication of the European Commission's White Paper on AI. With more than 130 participants, CEN-CENELEC/JTC 21 is one of the largest committees among all ESOs. Its core mission is to develop standards to ensure the development of safe and trustworthy AI systems aligned with EU values.

Following a formal request by the European Commission,¹⁷ CEN-CENELEC/JTC 21 is developing harmonised standards to support the implementation of the *EU AI Act*. These standards address a wide range of topics, including risk management and conformity assessment procedures, organisational governance, data quality, transparency, accuracy, robustness, and cybersecurity specifications for AI systems. They will define the technical specifications needed to meet the key requirements of the *EU AI Act* and will grant organisations that adopt them a presumption of conformity with the relevant legal provisions. Based on the European Commission's original standardisation request, these standards should have been adopted by 31 January 2025. The deadline was later postponed to August 2025 to accommodate for delays in the adoption of the *EU AI Act*. In Spring 2025, however, CEN-CENELEC announced that the standards supporting the implementation of the *EU AI Act* are further delayed and will likely be published in the summer of 2026.

Engagement in international standardisation

As stated by the European Commission, shaping international standards is a key priority for the EU and is in line with the goals of promoting global competitiveness and fostering international trade. The European Commission specifies the strategic objectives for international standardisation in its annual *Union work programme* and actively promotes the collaboration between ESOs and International Standardisation Bodies (ISBs) like IEC, ISO and ITU.

¹⁷ European Commission, 'Commission Implementing Decision on a Standardisation Request to the European Committee for Standardisation and the European Committee for Electrotechnical Standardisation in Support of Union Policy on Artificial Intelligence', May 2023, https://ec.europa.eu/growth/tools-databases/enorm/mandate/593_en.

For its part, the EU often transposes international standards into European standards when these sufficiently meet European needs and interests—one of the key roles of CEN-CENELEC/JTC 21 is to identify opportunities for alignment with international standards in the context of the *EU AI Act*. While this work is still ongoing, CEN-CENELEC/JTC 21 seems motivated to draw on existing standards wherever possible in order to meet the stringent deadline set in the standardisation request issued by the European Commission.

